

Smart QR-Based Attendance Management System

A Secure, Automated, and Practical Solution for College Environments



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Technology Stack
Spring Boot + React.js



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Problem Statement

Critical Issues in Current Attendance Management Systems



Time-Consuming Manual Process

Traditional roll call consumes 5-10 minutes per class, wasting valuable teaching time..



No Device or Network Verification

Systems lack validation of student device identity and network connection. Students can use multiple devices or connect from outside the campus network, making attendance tracking unreliable.



Static QR Code Vulnerability

Fixed QR codes can be easily photographed and shared via messaging apps. Students can mark attendance from outside the classroom, defeating the purpose of physical presence.



Unreliable Attendance Records

No audit trails or confidence scoring for attendance entries. Disputes arise when students claim they were present but not marked, with no way to verify authenticity.



Proxy Attendance Challenges

Friends can easily mark attendance for absent students. No mechanism to verify the actual identity of the person scanning the QR code, leading to widespread proxy attendance.



The Need: Secure & Practical Solution

A system that balances **security with usability**, specifically designed for college environments. Must prevent proxy attendance while remaining easy to use across diverse devices and technical skill levels.

Proposed Solution

Multi-Layered Security Architecture with Dynamic QR Codes



Time-Based Dynamic QR Codes

QR codes rotate every **20 seconds**, preventing screenshot sharing and proxy attendance. Each QR token contains a unique session identifier with automatic expiry.

20s

Expiry Time

100%

Unique Tokens



Web-Based System

No mobile app required – students use device browsers to access the web interface. Eliminates app installation barriers and works across all devices (Android, iOS, laptops).



Location Verification

GPS validation with **30m ideal, 100m maximum** distance. Soft-fail mechanism accounts for GPS drift indoors while maintaining location integrity.



Device Registration

Students explicitly register up to **5 devices**. New devices trigger a registration modal showing device slots used (X/5). Prevents device sharing and provides audit trail.

Device Management UI



Network Restriction

Attendance restricted to **college Wi-Fi network** IP ranges. Pre-scan connectivity check detects mobile data (4G/5G) and displays helpful error messages with troubleshooting steps.



Connect to College Wi-Fi and disable Mobile Data

Technology Stack & Architecture

Complete Technology Overview and System Workflow

Backend

 **Java + Spring Boot**
RESTful API, JWT Auth

Key Dependencies

Spring SecurityJWTJPA

Frontend

 **React.js**
Responsive Web UI

Key Libraries

QR ScannerGeolocationAxios

Database

 **MySQL**
Relational DB

Key Tables

usersattendance

System Workflow

- 1

Teacher Generates QR
Creates attendance session with 20s rotating QR
- 2

Student Scans QR
Web interface captures QR code and geolocation
- 3

Backend Validation
7-layer security pipeline validates request
- 4

Attendance Marked
Record saved with confidence level and audit trail

 **JWT Authentication**
Stateless, secure token-based auth for all API calls

HTTPS & Geolocation Setup

Development

Ngrok HTTPS Tunnel
ngrok http 3000 → HTTPS URL

Production

Let's Encrypt SSL
Domain: attendance.college.edu

⚠️ Geolocation requires HTTPS (or localhost)

Ready for Real-World Deployment

A Secure, Technically Feasible, and Practically Deployable Solution



Security Achievements

- ✓ **Dynamic QR codes** prevent screenshot sharing
- ✓ **7-layer validation** ensures comprehensive protection
- ✓ **Device registration** reduces proxy attendance



Usability Features

- ✓ **Web-based** eliminates app installation
- ✓ **GPS soft-fail** balances accuracy with usability
- ✓ **Clear error messages** guide troubleshooting

Thank You